

Nihar Prajapati

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Education

Pandit Deendayal Energy University (PDEU) 2023 – Present
B.Tech in Electronics and Communication Engineering
CGPA: 9.19

Research Experience

Undergraduate Researcher — FSO & THz Wireless Communication 2025 – Present
Under the supervision of Dr. N. Vishwakarma

- Conducting performance analysis of Free Space Optical (FSO) and Terahertz (THz) wireless communication systems.
- Modeling FSO/THz channel behavior under atmospheric impairments and propagation effects.
- Analyzing key performance metrics including BER, SNR, and ergodic capacity for FSO and THz links.

Undergraduate Researcher — TFET-Based Biosensor Design 2025 – Present
Under the supervision of Dr. Abhishek Ray

- Designing a double metal gate Tunnel FET (TFET) biosensor using Silvaco ATLAS with non-local BTBT modeling.
- Engineering dielectric cavity regions for biomolecule permittivity modulation to enhance sensitivity.
- Optimizing high- κ gate stack, source/channel doping, and subthreshold swing for ultra-low-power detection.

Projects

Military Pager System

- Designed a long-range secure communication system without GSM/Internet using LoRa modulation.
- Implemented base-station routing with acknowledgment and retransmission protocol for reliable delivery.
- Optimized the system for low-power operation and deployment in network-denied environments.

Real-Time Fixed-Point FFT Engine on ATmega2560

- Implemented a radix-2 FFT in Q-format fixed-point arithmetic on an 8-bit AVR without FPU/DSP.
- Developed cycle-accurate butterfly scheduling with overflow-safe scaling techniques.
- Integrated timer-driven ADC sampling and UART-based magnitude spectrum transmission.

8×1 Multiplexer: RTL to GDSII (GitHub)

- Designed and verified Verilog RTL and performed synthesis using Synopsys Design Compiler.
- Executed full physical design flow including floorplanning, placement, CTS, and routing in ICC2.
- Achieved timing closure and power analysis using Synopsys PrimeTime with positive slack.

FM Transmitter–Receiver System

- Designed a Colpitts oscillator with LC tank for carrier generation in the 88–108 MHz FM band.
- Implemented RF amplification using BC547B and analyzed frequency deviation and noise performance.
- Validated wireless transmission through antenna matching and practical receiver testing.

GroqShell on ESP32-S3 (GitHub)

- Built an embedded AI shell interfacing the LLaMA-3.1 model via the Groq Cloud API over Wi-Fi.
- Implemented TLS-secured REST communication, JSON parsing, and UART-based human–machine interface.
- Integrated FreeRTOS multitasking for concurrent networking, I/O handling, and response processing.

Certifications & Training

A Primer to Mathematical Optimization

NPTEL (SWAYAM), IIT (BHU) Varanasi — 12 Week Course (Jul–Oct 2025)

Score: 57% (Top 21 Certified)

Verified Certificate

Image Processing using Python Workshop — PDEU

- Medical imaging, CGAN-based MRI enhancement, ULDCT super-resolution.
- Morphological processing, transformation techniques.
- Runner-Up in problem-solving competition.

Leadership

Society of Mathematics, PDEU — Head of Research

- Delivered talks and research workshops on mathematical modeling.

IEEE Computer Society, PDEU — Technical Committee Member

- Organized technical seminars and mentoring sessions.

Skills

Programming: Python, C, C++, R, Arduino

DSP & Libraries: NumPy, Pandas, Matplotlib

Simulation: MATLAB, Simulink, Silvaco TCAD, GNU Radio

EDA: KiCad, Proteus, Synopsys DC, ICC2, PrimeTime

Communication: Wireshark, LoRa, GSM, SDR

Embedded: ESP32, ESP8266, Raspberry Pi, ATmega2560 (AVR)

OS & Tools: Linux, Ubuntu, Git, GitHub, ESP-IDF, VS Code

Volunteer Experience

December 2024 39th National Symposium on Plasma Science and Technology (PLASMA 2024)

- Volunteered in organizing and coordinating the national symposium with 500+ participants, including top scientists nationwide.
- Presented innovative projects and explored plasma models developed by IPR.
- Met Shri K. N. Vyas, Former Secretary, DAE, during the event.